

Assignment-06

1. Create a tuple with different data and create an empty tuple using built-in function.

```
t1 = (1, True, 2.4, "Silicon")
empt = tuple()

print("The mixed tuple is: ", t1)
print("Empty tuple is: ", empt)
```

The mixed tuple is: (1, True, 2.4, 'Silicon')

Empty tuple is: ()

2. Print the 3rd element from beginning and 4th element from end of a tuple.

```
a = (10, 20, 30.5, True, 50, 60, 70)
print("3rd element from beginning: ", a[2])
print("4th element from end: ", a[-4])
```

3rd element from beginning: 30.5

4th element from end: True

3. Count the number of items present in a tuple and then add two new items into it (one item using concatenation operator and another item using built-in function).

```
tup = (10, 20, 30)

count = len(tup)
print("Number of items in the tuple:", count)

tup = tup + (40,)
print("After concatenation:", tup)

temp = list(tup)
temp.append(50)
tup = tuple(temp)
print("After using built-in functions:", tup)
```

Number of items in the tuple: 3

After concatenation: (10, 20, 30, 40)

After using built-in functions: (10, 20, 30, 40, 50)

4. Find the number of occurrences for any repeated item present in a tuple.

```
t = (10, 20, 30, 20, 40, 20, 50, 30)
n = int(input("Enter the item: "))

print("The occurrence is: ", t.count(n))
```

Enter the item: 20
The occurrence is: 3

5. Display the elements present in a tuple which are at the even position.

```
t = (10, 20, 30, 40, 50, 60, 70, 80)

print("Elements at even position:")

for i in range(1, len(t), 2):
    print(t[i])
```

Elements at even position:
20
40
60
80

6. Remove the elements of a tuple which are present in the odd index position.

```
t = (10, 20, 30, 40, 50, 60, 70, 80)

t = tuple(t[i] for i in range(len(t)) if i % 2 == 0)

print("Tuple after removing elements at odd index position:",t)
```

Tuple after removing elements at odd index position: (10, 30, 50, 70)

7. Remove the odd number elements from a tuple.

```
t = (10, 15, 20, 25, 30, 35, 40, 45)

t = tuple(x for x in t if x % 2 == 0)

print("Tuple after removing odd numbers:",t)
```

Tuple after removing odd numbers: (10, 20, 30, 40)

8. Reverse the elements of a tuple by using and without using a built-in function.

```
t = (10, 20, 30, 40, 50)
t = tuple(reversed(t))
print("Reversed tuple:",t)
```

Reversed tuple: (50, 40, 30, 20, 10)

9. Sort the elements present in a tuple in descending order.

```
t = (10, 50, 20, 40, 30)
t = tuple(sorted(t, reverse=True))
print("Tuple in descending order:",t)
```

Tuple in descending order: (50, 40, 30, 20, 10)

10. Take a tuple of mixed data types and find the square of each element.

```
t = (2, 4.5, 6, 8.5, 10)
for i in t:
    print(i, "square =", i ** 2)
```

2 square = 4
4.5 square = 20.25
6 square = 36
8.5 square = 72.25
10 square = 100

11. Given few numbers in a tuple, return 'True' if the first and last numbers are same.

```
t = (10, 20, 30, 40, 10)
if t[0] == t[-1]:
    print(True)
else:
    print(False)
```

True

12. Print only those elements present in a tuple which are divisible by 5.

```
t = (10, 12, 15, 23, 25, 31, 40, 47)

print("Elements divisible by 5:")
for i in t:
    if i % 5 == 0:
        print(i)
```

```
Elements divisible by 5:
10
15
25
40
```

13. Given a tuple of numbers, create another tuple such that it contains only the even numbers of the first tuple.

```
t = (10, 15, 22, 35, 40, 47, 50, 63)

new_tuple = ()

for i in t:
    if i % 2 == 0:
        new_tuple = new_tuple + (i,)

print("Tuple containing even numbers:", new_tuple)
```

```
Tuple containing even numbers: (10, 22, 40, 50)
```

14. Display the largest number of a tuple without using the built-in function max(). The user needs to input the values in tuple from the keyboard.

```
a = int(input("Enter the no. of elements: "))
T1 = ()

for i in range(a):
    ele = int(input("Enter the elements: "))
    T1 = T1 + (ele,)

largest = T1[0]

for i in T1:
    if i > largest:
        largest = i

print(T1)
print("Largest element =", largest)
```

```
Enter the no. of elements: 5
Enter the elements: 25
Enter the elements: 10
Enter the elements: 45
```

Enter the elements: 30

Enter the elements: 15

(25, 10, 45, 30, 15)

Largest element = 45

15. Replace the last value of the tuple with 100. Sample list: [(1, 2, 3), (4, 5, 6), (7, 8, 9)] Output: [(1, 2, 100), (4, 5, 100), (7, 8, 100)]

```
T1 = [(1, 2, 3), (4, 5, 6), (7, 8, 9)]
T2 = []
```

```
for i in T1:
    T2.append(i[:-1] + (100,))
```

```
print(T2)
```

[(1, 2, 100), (4, 5, 100), (7, 8, 100)]

16. Sort a tuple by its float element. Sample data: [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')] Output: [('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]

```
T1 = [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')]
```

```
T1 = sorted(T1, reverse=True)
print(T1)
```

[('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]

17. Count the number of elements in a tuple until an element is a list.

```
T1 = (10, 20, 30, [40, 50], 60, 70)
count = 0
```

```
for i in T1:
    if type(i) == list:
        break
    count = count + 1
```

```
print("Number of elements =", count)
```

Number of elements = 3

18. Swap the two tuples: tuple1 = (11, 22) tuple2 = (99, 88) Output: tuple1 = (99, 88) tuple2 = (11, 22)

```
tuple1 = (11, 22)
tuple2 = (99, 88)

tuple1, tuple2 = tuple2, tuple1

print("tuple1 =", tuple1)
print("tuple2 =", tuple2)

tuple1 = (99, 88)
tuple2 = (11, 22)
```